

Integrating glycosylation in *de novo* protein design with ReGlyco Binder Design Filter

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Abstract

Artificial Intelligence (AI)-based methods for 3D protein structure prediction are revolutionising structural biology¹⁻⁷, providing novel templates for experimental data refinement and an on demand 3D perspective on any molecular architecture and protein-protein interaction (PPI). Regardless of the inherent limitations of the various approaches available to date, the continuous improvement of the algorithms, the broad availability of open access (OA) web servers^{3,8}, software packages^{6,9} and databases^{10,11} are bound to accelerate the discovery and optimization of novel biopharmaceuticals^{12,13}. Within this context, the development of computational pipelines for the *de novo* design of target-specific protein binders^{12,14,15} is especially exciting. As it stands, these processes are still rather inefficient¹⁶ and expensive, rapidly outputting thousands of designs relatively quickly, which translate into meagre yields. **Here we show how the explicit integration of glycosylation as a filter in the 3D *de novo* design pipeline can significantly improve efficiency and reduce laboratory costs with minimal additional computational resources.** As a proof-of-concept, we used the GlycoShape database and ReGlyco tools (<https://glycoshape.org>)¹⁷ to filter the results of a recent open competition launched by Adaptic Bio for the design of binders as inhibitors against the heavily glycosylated Nipah virus glycoprotein (NiV-G) (<https://proteinbase.com/competitions/adapticv-nipah-competition>). Screening of the 1,201 selected designs in block with ReGlyco, refined with the new ReGlyco Rotamer tool, flagged 11% of non-binders prior to experiment in approximately 3 hours on a dual-core CPU. We complement this analysis with a demo colab notebook (https://colab.research.google.com/github/Ojas-Singh/GlycoShape-Resources/blob/main/colab/ReGlyco_Binder_Design_filter.ipynb) to illustrate our workflow. In this demo users can design 'mini binders' against human erythropoietin (hEPO) by integrating GlycoShape resources with the RFDiffusion3 (RFD3) pipeline¹⁸ from the Institute for Protein Design (IDP).

Introduction.

Glycosylation is a sophisticated, enzymatically-driven^{19,20}, non-templated process where glycans are added co- and post-translationally to proteins and processed during trafficking along the secretory pathway. Integration of protein glycosylation in a 3D context is non-trivial because all glycoforms are heterogeneous, both in terms of glycan site occupancy (macroheterogeneity) and type of glycans at each site (microheterogeneity). Glycans come in all shapes and sizes^{21,22}, which are highly species-specific. Most secreted proteins and biologic therapeutics are glycosylated, where sugars can partially or completely occlude accessible surfaces, delimiting protein-protein interaction (PPI) to ligands, guiding multimerization and/or receptor docking, while affecting folding, stability, trafficking and pharmacokinetic properties of proteins^{23,24}. Current biomolecular design and structure prediction methods are protein-centric where post-translational modifications (PTMs), such

as glycosylation, are largely overlooked. While we can argue that AI-tools implicitly take into account the effects of PTMs on protein structure due to training²⁵, neglecting to include glycosylation explicitly in a 3D-based protein *de novo* design platform can lead to non-negligible numbers of false positives, needlessly increasing experimental efforts and laboratory costs.

Both AlphaFold 3 (AF3)³ and RosettaFold All-Atoms (RFAA)⁶ allow users to include glycosylation at different stages of the structure prediction process, yet with a limited repertoire of structures and low accuracy against experiment^{3,6}. This poor outcome should not be surprising, as the 3D structures of glycans available for AI-training are too few, too fragmented and seldom inaccurate²⁶, despite ongoing efforts in the area^{27,28}. Indeed, data from <https://glyconavi.org/> shows that only 6,873 proteins in the PDB contain N-glycans, which are the most abundant type in protein structured domains, for a total of 41,951 glycan structures. Of those 51% consist of a single monosaccharide, with less than 2,054 structures containing 7 or more monosaccharides.

GlycoShape is a structural database and toolbox¹⁷ designed to supplement the lack of 3D information on glycans with data obtained from molecular dynamics (MD) equilibrium simulations. To date GlycoShape is the largest and most diverse OA repository of 3D glycan structures worldwide, with 920 unique structures obtained by clustering analysis of approximately 2 ms of cumulative MD sampling of glycan structures largely from the human glycome, but with examples from all phyla. The 3D database is complemented by ReGlyco, a highly computationally efficient tool that selects glycans structures from the database and links them to the protein at the desired site by minimizing a loss function¹⁷. ReGlyco Ensemble is a recent expansion of ReGlyco that can be used to select multiple conformers (up to 500) from the weighted ensembles from our in-house MD trajectories²⁹, maintaining the distribution consistency given by the clustering analysis pipeline¹⁷ of the MD data.

To demonstrate how including glycosylation explicitly in the 3D *de novo* design of target-specific binders with GlycoShape can increase yields and reduce laboratory costs, we rescreened the results of the open competition recently launched by Adaptyv Bio (<https://www.adaptyvbio.com/>) for the design of binders against the Nipah virus glycoprotein (NiV-G) (<https://proteinbase.com/competitions/adaptyv-nipah-competition>) as potential inhibitors of viral entry, see **Figure 1**.

The Nipah virus (NiV) is a zoonotic henipavirus that can infect humans with an extremely high fatality rate, estimated by the World Health Organization (WHO) at 40% to 75% (<https://www.who.int/news-room/fact-sheets/detail/nipah-virus>). Outbreaks have been reported periodically since 1998 across south-east Asia, with the most recent one in India in 2026. There are currently no therapeutics or vaccines approved against NiV, while there is an urgent need for the development of strategies to block or to prevent infection. NiV infection requires fusion of the viral envelope with the host cell membrane. This process is mediated by the concerted action of two envelope glycoproteins, namely NiV-G³⁰ and the NiV fusion (NiV-F) glycoprotein³¹. NiV-G initiates entry by binding to the transmembrane protein tyrosine kinases ephrin-B2 or ephrin-B3 as receptors³²⁻³⁵, see **Figure 1.f**.

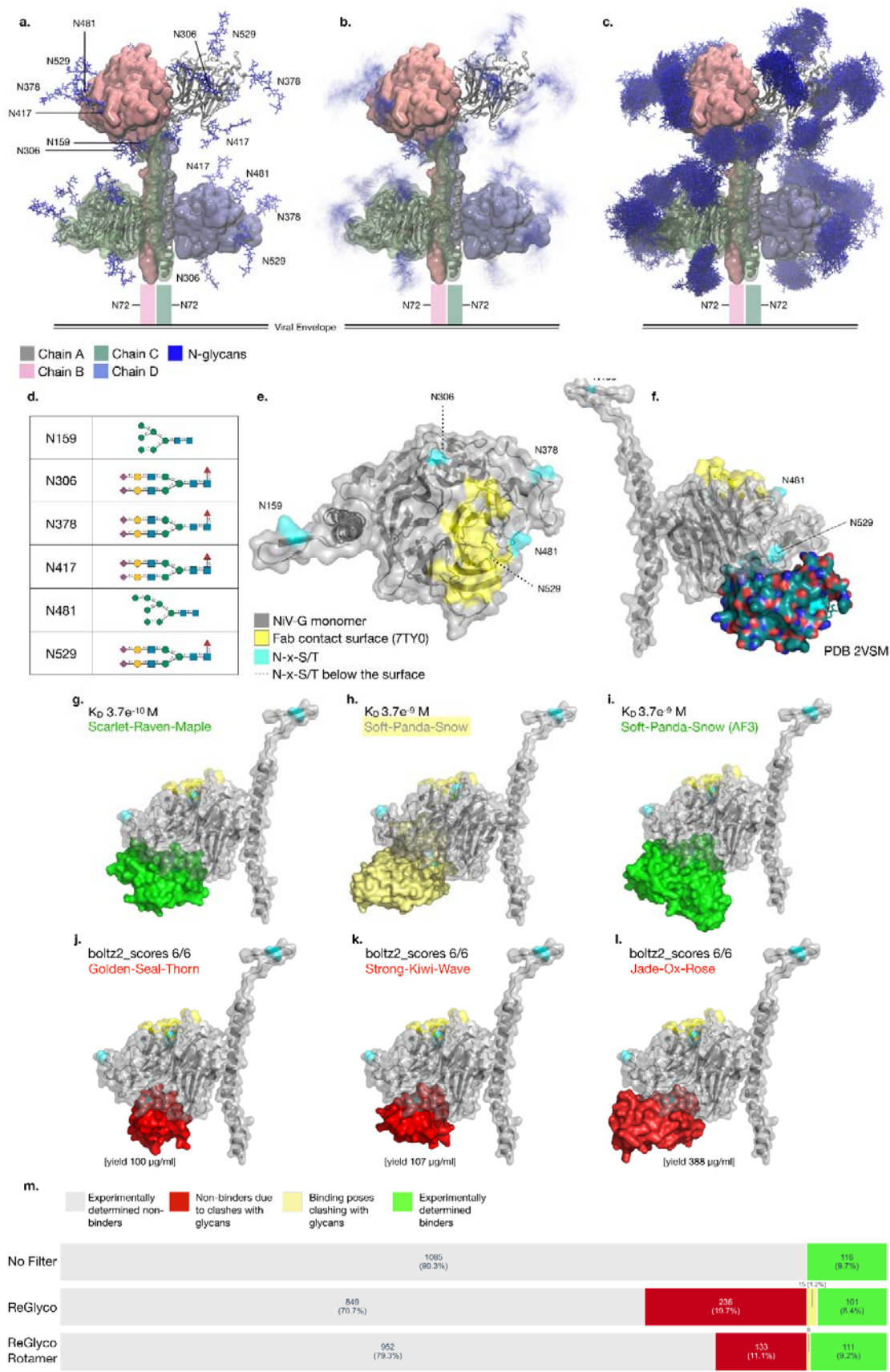


Figure 1. a. Structure of the Nipah Virus Glycoprotein (NiV-G) homotetramer (aa 92-602) reconstructed from cryo-EM structures (PDB 7TXZ and 7TY0) bound to broadly neutralising antibody nAH1.3 Fabs³⁰ (not shown). Static (single) glycans 3D structures (shown with sticks in blue) were reconstructed with GlycoShape ReGlyco¹⁷, where the choice of the glycan type was guided by glycoproteomics analysis³⁶. The visible six glycan sites on chains A, B and D are mapped onto the structure, while glycans on chain C are not labelled for clarity. **b.** Structure of the glycosylated NiV-G reconstructed with GlycoShape ReGlyco Ensemble²⁹ using 150 frames from the MD trajectories of each selected glycan. Rendering effect is obtained with a “long exposure” filter to reflect the glycan dynamics. Sterically constrained glycans and rigid regions of the saccharides can be identified by the darker blue colour. **c.** Structure of the glycosylated NiV-G as in panel b. but with each one of the 150 glycan frames shown in dark blue, to illustrate the potentially occluded volume of the NiV-G protein surface. **d.** Glycans selected at each site for the NiV-G reconstruction based on ref³⁶, represented with SNFG³⁷. GlyTouCan IDs are G54258NG (biantennary complex), G00028MO (oligomannose Man5), G97292GW (oligomannose Man7, chain A), G65518IK (oligomannose Man7, chain B), G00883ZT (oligomannose Man7, chains C and D). **e.** Close up view on one of the four NiV-G monomers from PDB 7TY0 (grey surface and cartoons) from the top, where the binding region occupied by the Fab is highlighted in yellow. Occupied sequons (N-x-S/T) are highlighted in cyan and labelled according to the PDB numbering. **f.** Close up view on one of the four NiV-G monomers from PDB 7TY0 (chain D, grey surface and cartoons) aligned to PDB 2VSM with bound ephrinB2 (surface with C atoms in teal, O in red, N in blue). **g.** From the Adapticv Bio binder design competition data released to date (1,201 designs), confirmed binders are shown in green, non-binders in red, binders but not in the submitted pose in yellow. Proteins are rendered as surface, translucent for the target (grey) and solid for the binder. Glycosylation sites are shown in cyan. Highest scoring binder id: Scarled-Raven-Maple (137 aa). **h.** 7th highest scoring binder id (submitted pose): Soft-Panda-Snow (149 aa). This pose failed the ReGlyco filter, clashing with glycan at N529. **i.** 7th highest scoring binder id: Soft-Panda-Snow (149 aa). Alternative pose obtained with AF3 (no glycans). **j.** All non-binders represented in panels j-l were selected from the pool based on a composite score from six structural metrics (*boltz2_complex_iplddt*, *shape_complimentarity_boltz2_binder_ss*, *boltz2_min_ipsae*, *boltz2_ipsae*, *boltz2_plddt*, *boltz2_lis*), alongside a binder-like pass count defined as the number of metrics exceeding the median value observed for confirmed binders. All designs shown were successfully expressed in *E. coli*. Non-binder id: Golden-Sea-Thorn (101 aa). This pose failed the ReGlyco filter, clashing with glycan at N306. **k.** Non-binder id: Strong-Kiwi-Wave (114 aa). This pose failed the ReGlyco filter, clashing with glycan at N306. **l.** Non-binder id: Jade-Ox-Rose (110 aa). This pose failed the ReGlyco filter, clashing with glycan at N306. **m.** Results of the Adapticv Bio binder design competition without ReGlyco filtering are shown in the top bar. The middle bar shows the effect of ReGlyco filtering in flagging designs for clashes with one or more glycosylation sites. The bottom bar indicates the effect of the refinement with ReGlyco Rotamer, which introduces flexibility in the Asn sidechain of the aglycon. Molecular rendering with VMD³⁸ (<https://www.ks.uiuc.edu/Research/vmd/>) and pymol (www.pymol.org).

In Oct 2025 Adapticv Bio (<https://www.adapticvbio.com/>) launched an open competition for the *de novo* design of binders targeting NiV-G as inhibitors of ephrin-B2 interaction, and thus based on the crystal structure of the NiV-G monomer in complex with ephrin-B2³⁴ (PDB 2VSM), see **Figure 1.f**. Of 10,000+ submissions of novel ephrin-B2-based binders, with entries within a range between 100 and 150 kDa, 1,201 designs were selected for expression in *E. coli*, with 1,028 successfully expressed (86%). Only 116 of the submitted entries selected were confirmed as binding NiV-G (10% success rate), see **Figure 1.m**. None of the entries included glycosylation explicitly of either the binder or of the target NiV-G. While the binders were expressed in *E. coli* for the binding assays, and thus were not glycosylated, the NiV-G was expressed in HEK293 cells and thus it was fully glycosylated, as shown in **Figures 1.a-c**.

To illustrate the extent of the glycosylation of the NiV-G targeted for design here, we reconstructed the glycans with ReGlyco based on the NiV-G cryo-EM structure³⁰ and produced a representative fully glycosylated tetramer based on recently published glycoproteomics data³⁶, see **Figure 1**. We used ReGlyco Ensemble to link the glycan structures to the protein scaffold using 150 frames from the MD trajectories for each chosen

glycan, see **Figures 1.b** and **1.c**. As a note of interest, the rendering in **Figure 1.b** is obtained with a bespoke filter we designed to mimic the effect of a long-exposure photograph. This perspective highlights the glycan fragments that are less dynamic because of inherent structural restraints, or protein surface complementarity, while blurring out the contributions from the most dynamic fragments, providing a potentially more realistic graphical representation of glycan shielding and screening relative to the timeframe overlay in **Figure 1.c**. The results of our filtering are presented and discussed below. As a complement to this case, we present an OA colab notebook demo to illustrate our workflow where users can run a 3D *de novo* design of mini-binders to the heavily glycosylated hEPO by integrating GlycoShape resources with RFdiffusion 3 (RF3) protein design pipeline¹⁸.

Filtering the NiV-G binding competition designs with ReGlyco and ReGlyco Rotamer

We downloaded and screened the 1,201 designs selected to date for expression by Adapticv Bio to filter potential clashes due to the NiV-G extensive glycosylation, see **Figure 1**. The 3D structures of all binders sequences submitted were predicted with Boltz-2⁹ for scoring and selection purposes. We used those 3D structures in our filtering pipeline. We first used the GlcNAc scanning mode¹⁷, currently set as the 'default' option in the reconstruction pipeline on ReGlyco (<https://glycoshape.org/reglyco>), to identify the glycosylation sites on the structure of the NiV-G monomer structure (PDB 2VSM), which was used as a 3D model of the receptor in the competition. GlcNAc scanning correctly identified six glycosylation sites³⁶ as potentially occupied, namely N159, N306, N378, N417, N481 and N529. We linked to all of these a Man₅-GlcNAc₂- oligomannose (GlyTouCan ID G000208MO) for screening purposes with two filtering approaches, 1) ReGlyco, where the glycan is linked to the aglycon Asn sidechains in the same conformation as the one predicted by Boltz-2, i.e. rigid rotamers, 2) ReGlyco Rotamer, a new tool we developed for this work that allows the aglycon Asn sidechain torsions to access different rotamers based on Dunbrack's Smooth Backbone Dependent Rotameric Library³⁹ (<https://dunbrack.fccc.edu/lab/bbdep2010>) to clear steric clashes that depend on the Asn sidechain orientation alone¹⁷.

The results of the competition show that only 116 designs out of the 1,201 that Adapticv Bio selected for protein expression in *E coli* were found positive for binding to NiV-G expressed in HEK293 cells, see **Figure 1.m**; 173 were not successfully expressed, and 917 did not bind NiV-G. Parsing the 1,201 binders designs in-block through the ReGlyco filter took 3 hours and 16 minutes on a dual core CPU. ReGlyco flagged 251 complexes as non-binders due to steric clashes involving at least one glycosylation site. Of the 251 designs flagged, 236 were among the confirmed non-binders by experiment and 15 were among the confirmed binders, see **Figure 1.m**. We refined the screening by enabling rotameric freedom through ReGlyco Rotamer, which reduced the number of 251 designs flagged by Re-Glyco for steric clashes to 138. Of those 133 were among the confirmed non-binders and 5 among the confirmed binders, see **Figure 1.m**. We visually assessed the clashes in these 5 entries and decided to rebuild the corresponding complexes with the AF3 server (<https://alphafoldserver.com/>) with no added glycans, to check for binding poses alternative to Boltz-2 predictions provided online. In all only 1 of those 5 cases, namely the 'Soft Panda Snow' ($K_D = 4 \mu M$) entry shown in **Figure 1.h** and **i**, AF3 produced a non-clashing pose, suggesting that the use of different prediction methods can in some instances help refine the output. All complexes from the Adapticv Bio Re-Glyco filtering analysis are available on <https://glycoshape.org/downloads/adapticv-nipah-competition>. As a note of interest, our first

attempt to build all 5 binders flagged by ReGlyco on the AF3 server with a fully glycosylated NiV-G monomer as a target, timed-out with no output after 24 hrs. We re-run those attempts more recently and each prediction took 4 hrs to terminate successfully. This indicates that the AF3 algorithm is continuously improving and it is a very useful prediction tool for complexes involving glycosylated biomolecules, yet extremely time consuming (to date) relative to the AI+ReGlyco scheme we presented here and discuss below within the framework of an OA demo platform for users to experiment on.

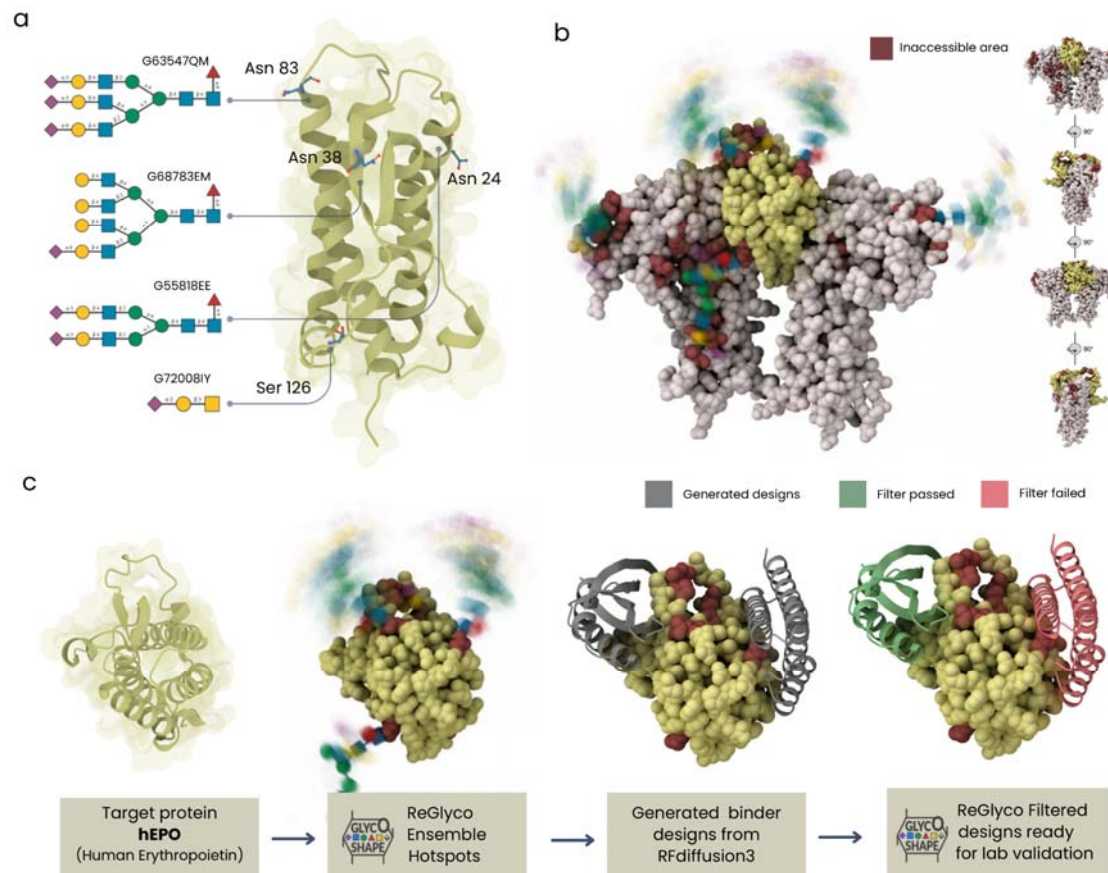


Figure 2.a. 3D structure of the human erythropoietin (hEPO) (from PDB 1EER) rendered in yellow cartoons with a transparent molecular surface. N-linked (3 sites) and O-linked (1 site) glycan sites are labelled according to the PDB numbering. The glycans selected for the protein reconstruction are shown in SNFG with GlyTouCan IDs. **b.** 3D structure of hEPO (yellow spheres) bound to the hEPO receptor (white spheres) (PDB 1EER). The exclusion surface areas due to glycosylation are highlighted with dark red spheres. Glycans are rendered in SNFG with the 'long exposure filter' to show their dynamics modulated by the protein surface at and around the site. On the right side of the panel the structure on the complex is progressively rotated over 360° in 90° steps around the z axis. **c.** Vignettes showing in four stages the RF3-ReGlyco mini-binder design workflow to hEPO as in the demo (https://colab.research.google.com/github/Ojas-Singh/GlycoShape-Resources/blob/main/colab/ReGlyco_Binder_Design_filter.ipynb). From the left-hand side, the 3D structure of hEPO is selected (yellow cartoons with a transparent molecular surface) as target protein. ReGlyco Ensemble is used to reconstruct the glycoprotein with the desired glycosylation. This process will output the inaccessible surface area due to the presence of glycans (dark red spheres) and the accessible surface area (yellow spheres) that can be targeted for binding, or 'hotspots'. RF3 is used to design mini-binders. Generated designs are shown in grey cartoons. These designs are filtered by ReGlyco to exclude designs clashing with one or more glycosylation sites: predicted binders are shown in green cartoons (filter passed) and predicted non-binders (filter failed) in red cartoons. Mol* (<https://molstar.org/>) was used for structure rendering.

A colab notebook for the design of binders to hEPO with RFDiffusion3 and ReGlyco

Human erythropoietin (hEPO) is a 30 kDa glycoprotein hormone that regulates the production of red-blood cells, or erythropoiesis, by binding a transmembrane dimeric receptor⁴⁰ (hEPOR), see **Figure 2**, located on the membrane of erythrocytes. hEPO is a heavily glycosylated biologic bearing three fully occupied N-glycosylation sites at N24, N38 and N83, and one O-glycosylation site at S126^{41,42}. Glycosylation, and more specifically sialylation, is a key determinant of hEPO pharmacokinetic properties, so that the hyperglycosylation of hEPO has been engineered successfully to produce a recombinant form of the hormone, known as NESP^{23,41}, with remarkably higher half-life. The extensive degree of hEPO glycosylation, and the position of the two additional sites in NESP, lead the hormone to bind hEPOR in designated (non-glycosylated) surface areas, see **Figure 2.b**. Here we illustrate our ReGlyco Binder Design Filter workflow through a demo colab notebook using hEPO as a sample target receptor. This choice is due primarily to hEPO low molecular weight compared to viral fusion proteins such as NiV-G and NiV-F discussed earlier, which are incompatible with the 16 GB GPU limit of the free notebook. Additionally, the glycoengineering of recombinant hormones and of small biologics is also of high interest⁴³⁻⁴⁵, as well as the promise and ambitions of *de novo* design technologies in this area.

The ReGlyco Binder Design Filter demo workflow is illustrated in **Figure 2.c** in four vignettes and explained step-by-step below. In the default mode ('quick' demo mode), the target receptor is hEPO (from PDB 1EER) with glycosylation selected at the known sites^{41,42} and glycan types, indicated by GlyTouCan⁴⁶ IDs, aligned with published glycomics profiles, see **Figure 2.a**. Users can leave the defaults for the bundled hEPO example, or switch to 'advanced' mode to upload a different target, provide their own glycan list, or change how the interaction patch is selected. Details on how to set up the 'advanced' mode are in the notebook, at the bottom of the workflow. We set a limit of up to 250 aa for the total length of the complex due to restrictions in the free version of the colab notebook resources. The user can select up to 6 residues as interaction hotspots and up to 20 mini-binders as outputs across all available hotspots regions. Each mini-binder is set up by default to a length of 80 aa, but this parameter can be changed to up to 130 aa. Three options are available to distribute the binders across the selected hotspots: 1) 'chunked' (default) to spread binders across different hotspots in the pool, 2) 'random' to randomly select different hotspots for each binder, and 3) 'rotate' to select hotspots in order as a target for each binder. Re-Glyco Ensemble is run to identify the surface areas of the receptor inaccessible due to the specific glycosylation selected, i.e. the 'glycosylation hotspots'. The output can be inspected visually in the Mol* GUI⁴⁷. At this point the demo version will generate a subset of 20 mini-binders using RFDiffusion3¹⁸ (RFD3) resources against the protein-only target. Each design candidate is set-up to target a different hotspot to produce a broad spread of designs that ReGlyco can triage downstream. The results can be viewed in the Mol* GUI⁴⁷. The designs produced are then filtered by ReGlyco for clashes with one or more glycosylation sites. This test can result in three outcomes, 1) 'pass', when no clashes are detected, 2) 'borderline', clashes are cleared by enabling rotameric degrees of freedom through ReGlyco Rotamer, and 3) 'fail' clash persists indicating serious overlap with one or more glycosylation sites. The demo default settings lead to 14 designs in the 'pass' category, 1 in the 'borderline', and 5 in the 'fail' due to irresolvable clashes with one or more glycosylation sites. All results with their respective classification label, can be viewed in the Mol* GUI⁴⁷.

Conclusions

In this work we have shown that glycosylation of the target receptor is a limiting factor to the spatial accessibility of the protein surface and for that reason it should be taken into account in *de novo* design processes. The screening of the results of the Adapticv Bio open competition for the design of inhibitors to the NiV entry and infection, highlights that ReGlyco and ReGlyco Rotamer can very rapidly and successfully flag non-binders prior to experimental testing. This step can be either by the users submitting the entries, and/or by those analysing the entries in block. We filtered all 1,201 entries with both ReGlyco and ReGlyco Rotamer and showed that this simple additional step can reduce laboratory effort and costs with minimal additional computing resources. On a dual-core CPU ReGlyco can successfully screen 1,201 complexes in approximately 3 hrs time and highlight that 20% (236) of those are non-binders because of steric clashes with glycans at at least one site. ReGlyco also flags as clashing 1.2% of actual binders, some of those among high affinity binders. Refinement with ReGlyco Rotamer flags 11% (133) of all structures as non-binders due to clashes with at least one glycosylation site, and clears most of the actual binders, with only 0.4% (5) left in the list. A careful analysis of those 5 complexes confirms that the clash is irresolvable and that the prediction of the binding pose may be incorrect. We have shown that changing the prediction algorithm provides alternative (non-clashing) in at least one case, suggesting that the use of multiple structure prediction tools could also improve the efficiency of the *de novo* design pipeline.

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Data Availability

Details on the ReGlyco Binder Design Filter pipeline and algorithms are provided in Supplementary Material. The ReGlyco Binder Design Filter colab notebook is available at https://colab.research.google.com/github/Ojas-Singh/GlycoShape-Resources/blob/main/colab/ReGlyco_Binder_Design_filter.ipynb The GlycoShape database and tools used in the colab notebook are available at <https://glycoshape.org>. The results of the ReGlyco filtering of the Adapticv Bio Open Competition are available on <https://glycoshape.org/downloads/adapticv-nipah-competition>.

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